Keywords in Each Notebook, in Simple Words

Oct 24, 2026 · @what are you

Each table lists the key words you meet in one notebook, with a meaning in plain language.

Notebook 01: Framing and provenance

Keyword	Meaning in simple words
Research question	The one clear question the project tries to answer
Provenance	Where the data came from and who made it
Licence (CC BY 4.0)	The rule for using the data: you may use it if you credit the authors
VLE	Virtual Learning Environment, the website where students study and click
Module	One course, for example module FFF
Presentation (course run)	One time a course is taught, named by year and start month (2014J = October 2014)
Unit of analysis	What one row of the table stands for: here, one student in one course run
Key (ID)	A column, or a group of columns, that points to exactly one row
Population	The group of students the results are about
Target (outcome)	The thing we want to predict: ending the course at risk
Conceptual map	A list that sorts each variable into a role: outcome, explanation, background or ID
SDG 4	UN Sustainable Development Goal 4: quality education for everyone

Notebook 02: The master table

Keyword	Meaning in simple words
Master table	The one clean table that all later notebooks use, with one row per student
Cutoff (prediction day)	The day we make the prediction (day 28); only earlier information is allowed
Early leaver	A student who left before day 28; removed because their result is already known
Data leakage	Using information the model would not have in real life, so results look too good
Aggregation	Adding many small records (every click) into one number per student
Click	One use of a page on the website; the notebook counts clicks by week and by type of material
Active days	The number of different days a student used the website
Banked score	A mark carried over from an earlier attempt, not work done in this course run
Missing value	A blank where a number should be; here the file writes it as `?`
Indicator (flag) column	A 0 or 1 column that records a fact, such as "deprivation band is missing"
Exact duplicate	Two rows that are identical in every column
Assertion	A check in the code that stops it if something that should be true is not
Transformation log	A list of every change made to the data, with how many rows it touched
Data dictionary	A table that explains each column: meaning, type and role

Notebook 03: Exploring and checking the data

Keyword	Meaning in simple words
Data-quality audit	A careful check of what is missing, wrong, odd or risky in the data
Missingness	How much data is missing, and whether the gaps follow a pattern
Outlier	A value far from most others, such as a student with 7,570 clicks
Skewness	How lopsided a distribution is; click counts have a long tail of heavy users
Histogram	A bar chart that shows how many students fall in each range of values
Rare category	A group with very few students, so its numbers are shaky
Near-zero variance	A column that is almost always the same value and so tells us nothing
Sensitivity check	Redoing a result a different way to see if the answer changes
Correlation (Spearman)	A number from -1 to 1 for how closely two things rise and fall together
Chi-square test	A test of whether two categories, such as deprivation and risk, are linked
Cramér's V	The strength of that link, from 0 (none) to 1 (perfect)
Effect size	How big a difference is; more useful than a bare yes or no from a test
Confidence interval	A range that likely contains the true value; narrow means more certain
Odds ratio	How many times higher the odds of trouble are in one group than another

Keyword	Meaning in simple words
Logistic regression	A simple model that predicts a yes or no outcome and shows each factor's push
Confounding	A third factor, such as the course, that makes two things look linked when they are not

Notebook 04: Building and choosing features

Keyword	Meaning in simple words
Feature	One fact about a student that the model uses
Feature engineering	Creating better facts from the raw data, for example turning clicks into a weekly trend
Feature selection	Choosing the few features that matter and dropping the rest
Percentile rank	Where a value sits among others, from 0 (lowest) to 1 (highest)
Relative feature	A fact compared with other students in the same course, not on its own
Log transform	Squeezing large numbers so a few huge values do not dominate
Capping	Cutting extreme values down to a fixed limit
Imputation	Filling in missing values with a sensible guess, such as the median
Standardising	Rescaling numbers so they have the same typical size
One-hot encoding	Turning a category into several 0/1 columns so a model can use it
Training and test split	Learn on 2013, then check on the unseen year 2014
Cross-validation	Practising on the training data by hiding a part, predicting it, and repeating

Keyword	Meaning in simple words
Multicollinearity (VIF)	Features that say nearly the same thing; the variance inflation factor measures it
Mutual information	A score for how much knowing a feature tells you about the outcome
Permutation importance	Shuffle one feature and see how much the model gets worse
Ablation	Adding or removing groups of features to see what each group contributes
Pipeline	A fixed chain of steps (fill, rescale, predict) that is run the same way every time

Notebook 05: Clustering

Keyword	Meaning in simple words
Clustering	Letting the computer put similar students into groups
Unsupervised learning	Learning from the data without being told the right answer
k-means	Picks k centres and puts each student with the nearest one
Hierarchical (Ward)	Starts with every student alone and keeps merging the most similar groups
DBSCAN	Finds crowded areas and calls students in empty areas "noise"; it found no clear crowds here
Gaussian mixture (GMM)	Assumes the data is a blend of bell-shaped groups and gives each student a chance of belonging to each
PCA	Squeezes many related columns into a few combined ones that keep most of the information

Keyword	Meaning in simple words
Silhouette score	How well separated the groups are; below 0.25 means weak structure
BIC	A score that rewards fit but penalises extra groups, used to choose how many groups
Bootstrap stability	Redo the grouping on resampled data and see if the groups stay the same
Adjusted Rand index (ARI)	How much two groupings agree, from 0 (chance) to 1 (identical)
z-score	How many typical steps a value is above or below average
Profile	A short description of what the students in one group do
Soft assignment	A student can be 70% in one group and 30% in another
Population stability index (PSI)	A number that says whether group sizes changed between years; under 0.1 means almost no change

Notebook 06: Rules and category tests

Keyword	Meaning in simple words
Item	One simple yes/no fact, such as "deprived" or "missed an assessment"
Association rule	A pattern "if A and B, then C" found in many students
Support	How many students the rule covers, as a share of everyone
Confidence	Of the students who match the "if" part, the share for whom the "then" part is true
Lift	How many times more often the outcome happens than usual; 1 means no effect

Keyword	Meaning in simple words
FP-growth	A fast method for finding all common combinations of items
Redundant rule	A longer rule that adds nothing over a simpler rule it contains
Multiple testing	When you try thousands of rules some look good by luck alone
Benjamini-Hochberg	A correction that keeps the share of false discoveries low
Contingency table	A table that counts students for every combination of two categories
Trend test	A test for whether risk rises or falls steadily across ordered levels
Mantel-Haenszel test	Compares groups inside each course, then combines the results, to remove the course effect
Breslow-Day test	Checks whether the effect is the same in every course
Correspondence analysis	A map that puts similar categories close together
Inertia	How much of the difference between categories a map dimension explains
Support list	A short ordered list of rules saying who to contact first

Notebook 07a: The prediction model, testing and fairness

Keyword	Meaning in simple words
Test set	Data kept hidden (2014) and used once at the end, like a final exam
Hyperparameter	A setting of the model chosen by you, such as how deep the trees may grow

Keyword	Meaning in simple words
Gradient boosting	A model that builds many small decision trees, each fixing the mistakes of the last
Primary model	The model chosen before the exam by a rule written in advance
AUC	A score from 0.5 (guessing) to 1.0 (perfect) for how well at-risk students are ranked above others
PR-AUC	A similar score that focuses on how many of the flagged students are really at risk
Calibration	Whether a predicted 40% really means about 40 in 100 students
Brier score	The average squared miss of the predicted probabilities; lower is better
Bootstrap interval	A range for a result found by resampling the data many times
Flagging policy	The rule for who to contact; here, the 30% with the highest predicted risk
Precision	Of the students flagged, the share who are really at risk
Recall	Of all at-risk students, the share who are flagged
False-positive rate	Of the students not at risk, the share flagged by mistake
Lift	How many times better than flagging at random
Fairness (recall gap)	The difference in recall between groups, such as deprived and affluent students
Baseline	A very simple method to beat, such as ranking by engagement alone
Interaction	A model term that lets an effect differ by group, such as by course
Replication	Checking that an earlier finding still holds in the new year

Keyword	Meaning in simple words
Cutoff sensitivity	Testing how accuracy changes if we predict earlier or later

Notebook 07b: Imbalance, leakage and drift

Keyword	Meaning in simple words
Class imbalance	One outcome is much rarer than the other, such as 17% Withdrawn
Class weights	Telling the model that mistakes on the rare group count more
Oversampling	Copying rare-group rows so the groups are the same size
Undersampling	Deleting common-group rows so the groups are the same size
SMOTE	Making new, made-up rare-group rows between real ones
Threshold tuning	Moving the cut-off that turns a probability into a yes or no
F1 score	One number that balances precision and recall
Future information	Facts from after the prediction day; using them is cheating
Target encoding	Replacing a category by its average outcome, which leaks if computed on all rows
Temporal split	Training on earlier years and testing on later ones, as in real life
Data drift	The data changes over time, so an old model can get worse
Covariate shift	The inputs change (new courses, new deadlines) but the link to the outcome stays

Keyword	Meaning in simple words
Concept drift	The link between inputs and outcome itself changes
KS statistic	The largest gap between two distributions; larger means a bigger change
Adversarial validation	Train a model to tell 2013 from 2014; if it can, the data has shifted
Monitoring plan	A list of what to check, how often, and when to act or retrain

Notebook 07c: Time series and A/B testing

Keyword	Meaning in simple words
Time series	Numbers recorded in time order, such as total clicks per day
Trend	The slow rise or fall of a series over time
Seasonality	A pattern that repeats, such as a weekly cycle
STL decomposition	Splitting a series into trend, repeating pattern and leftover
Autocorrelation	How much a series today looks like itself some days ago
Event study	Averaging what happens around the same kind of event, such as a deadline
Holdout	The part of the series hidden while the forecast is made, then used to score it
Naive forecast	Predict that tomorrow equals today
Seasonal naive forecast	Predict that next week repeats last week
Holt-Winters	A forecasting method that smooths the level, trend and weekly pattern
Analog (twin) forecast	Use the same course from last year as a template

Keyword	Meaning in simple words
MASE	Forecast error compared with a simple method; below 1 beats it
sMAPE	Forecast error as a percentage of the size of the numbers
Bias	Whether a forecast is usually too high or too low
A/B test	A fair experiment: random students get the help (A) or not (B) and results are compared
Randomisation	Letting chance decide who gets the help, so the groups are alike
Statistical power	The chance an experiment detects a real effect of a given size
Minimum detectable effect	The smallest change an experiment of this size can reliably see
Type I error	Finding an effect that is not there; kept at about 5%
Confounding by indication	Help goes to the sickest, so they look worse even if help works
Regression discontinuity	Compare students just above and just below the cut-off for help
Guardrail	A side measure watched to make sure the programme does no harm